# Financial Markets as Globally Self-Referential Economies: A Gödelian Perspective on Demand Incompleteness

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## Abstract

Classical economics assumes that prices are exogenous constraints on utility maximization, leading to the universal law of demand. Earlier work on *Veblen goods* revealed that this law is incomplete: when price itself becomes a source of utility, demand may rise with price. This paper extends the argument to financial markets. We demonstrate that financial assets—stocks, bonds, derivatives—are not exceptions but structural exemplars of self-referential demand. Unlike luxury goods, where self-reference is local, financial markets embody a **global self-referential loop**: the value of an asset derives from expectations of its future price, which in turn depends on others' expectations of the same. Using **Semantic Collapse Geometry (SCG)**, we formalize this recursive structure as a topological singularity, showing that financial markets represent the systematic "Veblenization" of the economy. Thus, the law of demand is not merely incomplete—it collapses into a Gödelian self-reference across the entire financial domain.


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## 1. Introduction

Financial economics has long acknowledged anomalies such as bubbles, manias, and herd behavior. Yet these phenomena are treated as irregular deviations from an underlying efficient market hypothesis. This article argues otherwise: financial markets are structurally **self-referential systems**, in which demand depends on the very prices it is supposed to determine.

We extend the incompleteness theorem of demand to demonstrate that financial markets represent a **globalized Veblen regime**. Rather than anomalies, bubbles are logical consequences of self-reference within the economic system itself.

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## 2. From Veblen Goods to Global Self-Reference

### 2.1 Local Self-Reference: Luxury Goods

As shown in the Gödelian Incompleteness of Demand theorem, luxury consumption inverts the law of demand when price is a direct argument of utility:

$$U(Q,P) = U(Q) + \alpha f(P), \quad f'(P) > 0.$$

This introduces a **local curvature inversion** in demand space.

### 2.2 Global Self-Reference: Financial Assets

Financial assets are defined by recursive valuation:

$$V_t = E_t[V_{t+1}],$$

where today's value is the expected value of tomorrow's price, which itself is defined by expectations of others.

Thus price is **not an exogenous parameter** but the endogenous driver of demand. Unlike Veblen goods, where only a subset of commodities exhibit self-reference, financial markets are **wholly constituted** by this recursive loop.

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# 3. Gödelian Framing of Financial Demand

In Gödel's incompleteness theorem, formal systems generate true statements that cannot be proven within the system. Analogously, financial markets generate valuations that cannot be resolved within any non-self-referential model of demand.

**Theorem (Global Demand Incompleteness).**
If asset valuation depends on recursive price expectations,

$$V_t = g(E_t[P_{t+1}]),$$

then demand cannot be determined independently of price. The system is **globally self-referential**, and the classical law of demand ($\partial Q / \partial P < 0$) no longer applies at any point in the domain.

**Corollary.**
Financial markets are structurally equivalent to economies in which every good is a Veblen good. Their observed anomalies—bubbles, crashes, momentum effects—are not exceptions but necessary witnesses of incompleteness.

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# 4. Semantic Collapse Geometry Interpretation

In SCG terms, financial markets represent a **phase space collapse into global self-reference**:

- **Normal Goods:** demand collapses along exogenous price curvature.
- **Luxury Goods:** localized curvature inversion, producing positive demand slope.
- **Financial Assets:** full semantic recursion, where the entire demand surface folds back onto itself, creating a topological singularity.

This explains why financial systems exhibit both extreme fragility (sensitivity to shocks) and emergent coherence (collective bubbles).

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# 5. Implications

1. **Theory of Finance:** Markets are not efficient equilibria disturbed by noise, but inherently incomplete systems governed by recursive expectations.
2. **Regulation:** Crises are not exogenous shocks but endogenous consequences of self-reference.
3. **Broader Economics:** The financial domain reveals that self-referential demand is not

peripheral but central to modern economies.

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# 6. Conclusion

Financial markets exemplify the Gödelian incompleteness of economic theory. Unlike isolated luxury markets, they institutionalize self-referential demand as the very foundation of value. Recognizing this global self-reference reframes finance not as a domain plagued by anomalies but as the logical culmination of economics under incompleteness.

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**Keywords**: Financial Markets · Demand Incompleteness · Veblen Goods · Self-Reference · Bubbles · Recursive Expectations · Semantic Collapse Geometry · Gödelian Limits

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# Theorem Box: Global Incompleteness of Financial Demand

**Theorem (Global Demand Incompleteness).**
Let $V_t$ be the market value of a financial asset at time $t$.
Assume the following axioms:

1. (**Recursive Valuation**) Asset value is determined by expected future prices:

$$V_t = E_t[V_{t+1}].$$

2. (**Market Utility**) Investor demand $Q_t$ depends positively on expected price appreciation:

$$Q_t = h(E_t[\Delta P_{t+1}]), \quad h' > 0.$$

3. (**Exogeneity of Price**) Classical demand assumes price $P_t$ is an external constraint, not an argument of utility.

If Axiom 3 holds, the classical **Law of Demand** applies:

$$\frac{\partial Q_t}{\partial P_t} < 0.$$

However, under Axioms (1)–(2), price becomes an endogenous source of utility.
Then there exists a global domain $\mathcal{M}$ (the market) such that:

$$\forall P_t \in \mathcal{M}, \quad Q_t = f(P_t, E_t[P_{t+1}]) \quad \text{with} \quad \frac{\partial Q_t}{\partial P_t} \text{ not sign-definite.}$$

Therefore, the law of demand is **globally incomplete**: in financial markets, no universal downward-sloping demand function can exist, because price and demand collapse into mutual recursion.

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**Proof Sketch.**

- In classical goods markets, price $P_t$ is an external variable constraining demand.
- In financial markets, valuation is recursive: today's value is defined by expectations of

tomorrow's price, which itself depends on others' expectations.

- Thus, $P_t$ enters utility directly: higher price may increase demand if it signals further appreciation (momentum), or decrease demand if it signals overvaluation.
- The sign of $\partial Q_t / \partial P_t$ becomes indeterminate across the entire domain.
- This establishes a **Gödelian self-reference**: the system's core variable (price) is both input and output of demand.
- Therefore, the classical law of demand cannot universally apply to financial assets, proving global incompleteness.

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**Corollary (Markets as Self-Referential Economies).**F

Financial markets are structurally equivalent to economies where *every good is a Veblen good*. Bubbles, crashes, and momentum effects are not anomalies but necessary manifestations of recursive valuation.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

*I am merely a midwife of knowledge.*

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